

Interdisciplinarity Formula Cheatsheet

Diversity and disparity

- **Rao–Stirling diversity** (Δ ; §§3.1, 4.1)

$$\Delta = \sum_{i \neq j} d_{ij} p_i p_j = 1 - \sum_{i,j} s_{ij} p_i p_j$$

with $d_{ij} = 1 - s_{ij}$ from the category similarity matrix.

- **Hill-type true diversity** (${}^q D^S$; §3.1)

$${}^q D^S = \left(\sum_{i=1}^N p_i \left(\sum_{j=1}^N s_{ij} p_j \right)^{q-1} \right)^{1/(1-q)}$$

similarity-sensitive Hill number with sensitivity parameter q .

- **Quadratic true diversity** (${}^2 D^S$; §3.1)

$${}^2 D^S = \frac{1}{\sum_{i,j} s_{ij} p_i p_j}$$

effective-number interpretation; related to Rao–Stirling via ${}^2 D^S = 1/(1 - \Delta)$.

- **Herfindahl concentration (toy example, §4.2)**

$$H_A = 0.300^2 + 0.075^2 + 0.175^2 + 0.350^2 + 0.100^2 = 0.258750$$

sample computation for Researcher A's portfolio.

Coherence and network structure

- **Bibliographic coupling similarity** (s_{ab} ; §3.2)

$$s_{ab} = \frac{\mathbf{r}_a \cdot \mathbf{r}_b}{\|\mathbf{r}_a\| \|\mathbf{r}_b\|}$$

Salton cosine between reference vectors $\mathbf{r}_a$ and $\mathbf{r}_b$.

- **Mean linkage strength** (S ; §§3.2, 4.1, robustness)

$$S = \frac{2}{N(N-1)} \sum_{a < b} s_{ab}$$

$$S = \frac{1}{\binom{n}{2}} \sum_{k < l} \cos(\mathbf{r}_k, \mathbf{r}_l) = \frac{1}{\binom{n}{2}} \sum_{k < l} \frac{\mathbf{r}_k \cdot \mathbf{r}_l}{\|\mathbf{r}_k\| \|\mathbf{r}_l\|}$$

average bibliographic-coupling strength across a publication set.

- **Betweenness centrality** (g_i ; §3.2)

$$g_i = \sum_{\substack{j,k \\ j \neq k \neq i}} \frac{g_{ijk}}{g_{jk}}$$

Freeman betweenness for positional bridging in journal networks.

- **Citing-distribution coherence** (C ; §3.3)

$$C = \sum_{i \neq j} p_{ij} d_{ij}$$

average cognitive distance among co-occurring citation categories.

- **DIV indicator (cited direction; §3.3)**

$$\text{DIV}_c = \frac{n_c}{N} (1 - G_c) \frac{\sum_{i \neq j} d_{ij}}{n_c(n_c - 1)}$$

variety–balance–disparity decomposition for citations received.

- **Gini coefficient for balance (G_c ; §3.3)**

$$G = \frac{\sum_{i=1}^n (2i - n - 1) x_i}{n \sum_{i=1}^n x_i}$$

applied to citation-share vector x sorted in non-decreasing order.

- **Cross-field effect proxy (E ; §4.1)**

$$E = \frac{\text{citations from articles whose primary category} \neq k^*}{\text{total citations received}}$$

share of citations arriving from outside each paper’s primary category.

Novelty measures

- **Generic divergence framing (§3.4)**

$$\nabla = f(|p(x) - p(E)|).$$

- **Probabilistic Jaccard divergence (∇_{PJ} ; §3.4)**

$$\nabla_{\text{PJ}}(x) = 1 - \frac{\sum_i \min(p_i(x), p_i(E))}{\sum_i \max(p_i(x), p_i(E))}.$$

- **Hellinger distance (∇_{Hel} ; §3.4)**

$$\nabla_{\text{Hel}}(x) = \frac{1}{\sqrt{2}} \sqrt{\sum_i \left(\sqrt{p_i(x)} - \sqrt{p_i(E)} \right)^2}.$$

- **Uzzi atypicality z-score (∇_{Uzzi} ; §3.4)**

$$\nabla_{\text{Uzzi}}(p_{i,j}, x) = \frac{p_{i,j}(x) - e_{i,j}(x)}{\sigma[e_{i,j}(x)]}, \quad p_{i,j}(x) > 0.$$

- **Timed novelty (§3.4)**

$$\text{Novelty}(x) = \sum_{i,j} t_0(i, j) \cdot [1 - z(i, j)],$$

weighting first-seen combinations by disparity.

Similarity-perturbation robustness (diversity; §4.3)

- **Uniform perturbation effect on Δ**

$$\Delta_{\text{new}} = \Delta_{\text{old}} - \varepsilon (1 - H)$$

with $H = \sum_i p_i^2$.

- **Gap evolution under perturbation**

$$\Delta_{A,\text{new}} - \Delta_{B,\text{new}} = (\Delta_A - \Delta_B) + \varepsilon (H_A - H_B)$$

vanishes at

$$\varepsilon^* = -\frac{\Delta_A - \Delta_B}{H_A - H_B}, \quad \varepsilon^* = \frac{0.020625}{0.058750} = 0.35106.$$

- **A/B vs. specialist separation**

$$\overline{\Delta}_{AB} - \Delta_C = \overline{\Delta}_{AB,0} - \Delta_{C,0} - \varepsilon (1 - \overline{H}_{AB} - (1 - H_C)).$$